



Metabolomics study and meta-analysis on the association between maternal pesticide exposome and birth outcomes

Xu Yang^{a,b,1}, Mingzhi Zhang^{a,b,1}, Ting Lu^{a,b}, Shiyao Chen^{a,b}, Xian Sun^{a,b}, Yusheng Guan^{a,b}, Yiyun Zhang^{a,b}, Ting Zhang^c, Rongli Sun^d, Bo Hang^e, Xinru Wang^{a,b}, Minjian Chen^{a,b,***}, Ying Chen^{c,**}, Yankai Xia^{a,b,*}

^a State Key Laboratory of Reproductive Medicine, School of Public Health, Nanjing Medical University, Nanjing, 211166, China

^b Key Laboratory of Modern Toxicology of Ministry of Education, School of Public Health, Nanjing Medical University, Nanjing, 211166, China

^c Wuxi Maternal and Child Health Hospital Affiliated to Nanjing Medical University, Wuxi, 214002, China

^d Key Laboratory of Environmental Medicine Engineering, Ministry of Education, School of Public Health, Southeast University, Nanjing, 210009, China

^e Biological Systems and Engineering Division, Lawrence Berkeley National Laboratory, Berkeley, CA, 94720, USA

ARTICLE INFO

Keywords:

Pesticides
Birth outcomes
Mecarbam
Hexachlorocyclohexane
Metabolomics

ABSTRACT

Background: Pregnant women are exposed to a number of pesticides which are widely used in China. Their potential risks on reproduction and infants are still unknown.

Objective: We aimed to investigate whether infant's birth weight and length of gestation were associated with levels of various pesticides in maternal blood based on Nanjing Medical University (NMU) affiliated hospitals data and meta-analysis, and also to explore the possible intermediate metabolomics pathways.

Methods: Eligible subjects ($n = 102$) were included in this study from the affiliated hospitals of NMU. Gas chromatography tandem mass spectrometry (GC/MS) and Q-Exactive mass spectrometer (QE) were used to detect 37 pesticides (9 organophosphorus pesticides, 7 organochlorine pesticides, 5 carbamate pesticides, and 16 others) and 161 metabolites (53 in amino acid metabolism, 47 in lipid metabolism, 18 in carbohydrate metabolism, 14 in nucleotide metabolism and 29 in other metabolisms) in maternal blood, respectively. Multi-linear regression and Bayesian kernel machine regression (BKMR) were performed to identify the association of single/mixed pesticide exposure in maternal blood with birth weight and length of gestation. Moreover, we conducted a meta-analysis including additional 2497 subjects to evaluate whether exposure to key pesticide, β -hexachlorocyclohexane (β -HCH) was associated with decreased birth weight globally. Mediation analysis was used to explore the metabolic alteration mediating the association between key pesticide exposure and birth outcomes.

Results: We found that decreased birth weight was significantly associated with increasing levels of mecarbam and β -HCH. We did not find any association between length of gestation and these pesticides. Among pesticides with detection rate more than 50%, BKMR analysis found an overall negative association of mixed pesticides exposure with birth weight, and verified that β -HCH was the key pesticide for such effect. Meta-analysis revealed a significantly negative association between exposure to β -HCH and birth weight. Metabolomics identified three metabolites and five metabolites as significant mediators for the effect of mecarbam and β -HCH, respectively, among which glyceraldehyde and its related glycerolipid metabolism and thyroxine and its related thyroid hormone metabolism were found to be the mostly enriched mediating metabolic pathway.

Conclusions: Based on the comprehensive pesticide exposome and metabolome wide associational study combined with meta-analysis, we found that prenatal exposure to β -HCH and mecarbam decreased birth weight via disrupting thyroid hormone metabolism and glyceraldehyde metabolism, providing new insights into the toxic effects of exposure to pesticides on birth outcomes.

* Corresponding author. State Key Laboratory of Reproductive Medicine, Center for Global Health, School of Public Health, Nanjing Medical University, No. 101 Longmian Road, 211166, Nanjing, China.

** Corresponding author. Central Lab, Wuxi Maternal and Child Health Hospital, Nanjing Medical University, 48 Huaishu Lane, Wuxi, 214002, China.

*** Corresponding author. State Key Laboratory of Reproductive Medicine, Center for Global Health, School of Public Health, Nanjing Medical University, No. 101 Longmian Road, 211166, Nanjing, China.

E-mail addresses: minjianchen@njmu.edu.cn (M. Chen), yingchen8@163.com (Y. Chen), yankaixia@njmu.edu.cn (Y. Xia).

¹ Xu Yang and Mingzhi Zhang contributed equally to this study and should be regarded as joint first authors.

1. Introduction

Pesticides are chemicals that are widely used around the world. World pesticide usage at the producer level reached nearly 6 billion pounds annually in both 2011 and 2012, and approximately 1.1 billion pounds of pesticides were used annually in the United States alone (EPA, 2017). Notably, China is the world's largest pesticide producer and consumer, and the total amount of pesticides used per year had reached 1.8 million tons in 2013 (Shuqin and Fang, 2018). Moreover, China has the largest population in the world, emphasizing the importance of the investigation into the health effect of pesticides in China.

In utero exposure may make people quite susceptible to chemical exposure. And as a result of the heavy use of pesticides in agriculture, pregnant women are at risk of being exposed to pesticides through water, fruits and vegetables. There is evidence that pesticides are able to cross the placenta (Perera et al., 2003), and they have been detected in amniotic fluid, indicating non-negligible fetal exposures (Bradman et al., 2003; Luzardo et al., 2009). Birth weight and length of gestation are important predictors of neonatal and infant health, and also related to maternal health. In the long run, abnormal birth weight is associated with incidence of coronary heart disease (CHD) and stroke (Lawlor et al., 2005), even affects the five-year CHD survival (Best et al., 2017). An increasing number of studies have investigated the association between pesticide exposure and birth outcomes including birth weight (Rauch et al., 2012; Guo et al., 2014; Toichuev et al., 2018) and length of gestation (Rauch et al., 2012). However, these studies focused on limited types or number of pesticides (Whyatt et al., 2004). There are various kinds of pesticides such as organophosphorus, organochlorine, and carbamate, and therefore it is urgent to examine the association between a large scale of pesticides exposure, namely pesticide exposure and birth outcomes.

Metabolomics analysis has been used in the studies on the mechanism underlying chemical toxicities and diseases (Bracewell-Milnes et al., 2017). It has also been introduced in the study on birth outcomes including birth weight and length of gestation (Favretto et al., 2012; Barberini et al., 2014; Robinson et al., 2018), and studies have proved the key role of metabolism in altered birth outcomes caused by prenatal chemical exposure (Tang et al., 2015).

The present study analyzed the relationship between the exposure levels of 37 major pesticides (pesticide exposome, covering organophosphorus, organochlorine, and carbamate) and birth outcomes. The pesticide selection was based on the usage, toxicity, and method availability. Mediation analysis was performed to understand what metabolism mediates the identified associations from the overall metabolomics perspective. Given the discrepancy on the association between maternal pesticide exposure and birth outcomes, for those significant results which can be merged with data reported in previous papers, we conducted a meta-analysis on the association between β -HCH and birth weight to address the current discrepancy and provided a global wide estimate of prenatal β -HCH effect on birth weight with a large population.

2. Methods and materials

2.1. Participants and recruitment

From September 2010 to October 2011, healthy pregnant women were randomly recruited from the affiliated hospitals of Nanjing Medical University (NMU). The mothers reported no assisted reproduction and gestational diseases. Newborn with genetic abnormalities or malformations were further excluded. A total of 102 women met the eligible criteria (response rate: 97.1%). The sample size estimation was calculated using G*Power software (Version 3.1.9) (Faul et al., 2007). The sample size for multiple linear regression was estimated with the inclusion of 9 covariates according to our statistical model.

The estimated sample sizes were 89 subjects to achieve 95% power with an alpha level of 0.05 and an effect size of 0.15. A total of 102 pregnant women were recruited in this study, which was also similar to a previous report on the same topic (Wang et al., 2012). All participants claimed that their life styles and environments had not changed for several months leading up to sample collection. No participant reported occupational exposure to pesticides for herself and her spouse. The study was approved by the Institutional Review Board (IRB) of Nanjing Medical University. All participants provided written informed consent prior to the study, and they provided adequate fasting blood samples during hospital admission for delivery which were stored at -80°C .

2.2. Outcome and covariate assessment

Information on newborns was obtained from the hospital maternity records by medical personnel, including infant sex, parity, and birth weight. Length of gestation was assigned based on self-reported date of last menstrual period (LMP). If there was a discordance between the clinical estimate and LMP (> 2 weeks), the first clinical estimation of length of gestation was used (Cheng et al., 2007). Other information such as age, height, family income, maternal education and lifestyle, was gathered during the baseline interview. Pre-pregnancy BMI was calculated as pre-pregnancy weight (kg) divided by the height (meter) squared.

2.3. Pesticides internal exposure assessment

The method for detecting pesticides was basically according to a previous report (Dana et al., 2002). An aliquot of 200 μL unknown blood serum samples were gently mixed with 10 μL the isotope labeled internal standard solution containing Alachlor- d_{13} , Chlorpyrifos- d_{10} and Dimethoate- d_6 , and then were mixed with 200 μL saturated ammonium sulfate and vortexed for 30 s. The supernatant was extracted after centrifuged (36,670 g) for 20 min, and was subjected to ProElut C18 SPE cartridges (200mg/3 mL, Dikma Technology, China) for cleaning and preconcentration. A triple quadrupole TSQ 8000 mass spectrometer (Thermo Fisher Scientific Inc. USA) was used to detect 37 pesticides including 9 organophosphorus pesticides, 7 organochlorine pesticides, 5 carbamate pesticides, and 16 others after separation by gas chromatography (Thermo Fisher Scientific, USA) with a column (TR-PESTICIDE II, 30 m, 0.25 mm, 0.25 mm, Thermo Fisher Scientific, USA). The limits of detection (LODs) ranged from 0.015 to 2.7 ng/mL (Table S1). The recoveries ranged from 80% to 120%. All pesticides showed intra-day and inter-day relative standard deviations (RSDs) less than 20%. Additional information on pesticide determination can be found in the supplemental methods.

2.4. The search strategy for meta-analysis

We searched for candidate studies on the databases of PubMed, Web of Science, Embase, and Cochrane Library before December 2019. The keywords were “organochlorine pesticide”, “hexachlorocyclohexane (HCH)”, “birth weight”, “birth outcome” and “pregnancy outcome”. The complete search strings are reported in Table S2. The inclusion criteria were as follows: 1) The exposure of interest was β -HCH; 2) The measurement of the exposure of interest was conducted by the analysis of blood not by the estimation through questionnaires; 3) The study examined the association between β -HCH exposure and birth weight; 4) the study reported regression coefficients with 95% CIs to extrapolate values. The quality of studies was assessed using the Newcastle-Ottawa Scale (NOS) (Stang, 2010). Two reviewers performed the data extraction and quality assessment. Disagreements were resolved through discussion until consensus was reached. This meta-analysis had been conducted according to the PRISMA checklists and followed the guideline to report the results (David Moher et al., 2015).

2.5. Metabolomics analysis

The metabolomic analysis was conducted using blood serum samples according to our previous reports (Zhou et al., 2018). First, an aliquot of 20 μ L blood serum sample was mixed with methanol in the volume ratio of 1:3. The samples were then centrifuged at 15,000 g for 15 min at 4 °C, and the supernatant was evaporated to dryness. An aliquot of 10 μ L of reconstitution was injected for the instrumental analysis. Ultra performance lipid chromatography tandem high resolution mass spectrometry (UPLC-HRMS) analysis was performed on an UPLC Ultimate 3000 system (Dionex), coupled with a Q-Exactive mass spectrometer (Thermo Fisher Scientific, Bremen, Germany). The instrument operated at a 70,000 resolution after full-scan acquisition, ranging from 70 to 1050 m/z. A multistep gradient had a mobile phase A consisting of 0.1% formic acid in ultra-pure water and a mobile phase B consisting of acetonitrile acidified with 0.1% formic acid; the gradient operated at a flow rate of 0.4 mL/min over a run time of 15 min. All samples were analyzed randomly to avoid bias from the injection order. Metabolite identification was based on the accurate mass and the retention time compared with the metabolite standards.

2.6. Statistical analysis

We used multiple linear regression models to examine associations between each of pesticides levels with birth weight and length of gestation. Pesticide levels in blood samples were all transformed using the natural logarithm (ln) before statistical analyses to normalize positively skewed distributions (Jusko, van den Dries et al., 2019). And pesticides levels below the LOD were given the value of the LOD/2 (Hornung and Reed, 1990). For multiple linear regression analysis, we selected a set of covariates based on both statistical and biological considerations. The final model's covariates for birth weight included risk factors associated with the outcomes at P -values ≤ 0.20 or risk factors known to be strongly associated with the outcomes from literature (Rauch et al., 2012), including continuous variables for pre-pregnancy BMI, maternal age, weight gain, length of gestation, as well as categorical variables for infant sex, passive-smoking (yes vs. no), drinking (yes vs. no), family annual income (< 200 thousand yuan vs. ≥ 200 thousand yuan) and maternal education ($< \text{Bachelor degree}$ vs. $\geq \text{Bachelor degree}$). Because the number of women smoking was relatively low ($n = 1$), smoking was not included in the models. At the same time, the final model's covariates for length of gestation included drinking, maternal age, passive-smoking and family income, maternal education and infant sex. For the maternal serum concentrations of each pesticide, those with a detection rate of $> 20\%$ and $< 50\%$ were stratified into two groups by using the LOD as the cut point. The levels of pesticides with detection rates above 50% were stratified into three groups by tertile. The sensitivity analyses were conducted with the adjustment of different sets of confounding factors. In the sensitivity analysis, pesticides with detection rate $> 50\%$ were treated as continuous variables and those with detection rate $< 50\%$ were treated as dichotomous variables.

For pesticides with detection rate $> 50\%$, we used Bayesian kernel machine regression (BKMR) models to evaluate the joint effect of mixed pesticides exposure on birth weight and conduct the interaction analysis. We conducted a variable selection method with 10,000 iterations by a Markov chain Monte Carlo algorithm (Bobb et al., 2015).

For meta-analysis, the effect estimates considered for pooling the data were beta regression coefficient (β) for continuous birth weight. Heterogeneity was assessed with the I^2 statistic (Julian PT Higgins, 2011). A random effects model was applied to merge the included studies for further analyses if the substantial heterogeneity was considered significant ($P < 0.1$ or $I^2 \geq 50\%$). Otherwise, we performed a summary estimate in a fixed-effects model. Potential sources of heterogeneity were explored, including clinical, methodological and statistical heterogeneity (Melsen et al., 2014). Potential publication bias was evaluated by funnel plots from Egger's test and Beggs test (Julian

PT Higgins, 2011). Sensitivity analyses using the "leave-one-out" approach were applied to assess whether any of the included studies could affect the robustness of current study.

We performed multiple linear regressions to assess the associations between selected pesticides and metabolomics data, and between selected metabolites and birth weight. All models were adjusted by pre-pregnancy BMI, maternal age, weight gain, length of gestation, as well as infant sex, passive-smoking, drinking, family annual income and maternal education. To further estimate the proportion of the β -HCH exposure effect on birth weight as mediated by the metabolites, we used mediation analysis to investigate possible associations between pesticides and birth outcomes by examining how they were related to a third intermediate variable, the blood serum metabolite. We accomplished this by dividing the total effect into direct effect and indirect effect. The direct effect can be interpreted as the treatment effect on the outcome when the mediator variable is at a fixed level. The indirect effect can be interpreted as effects on the outcome of changes of the exposure that act through mediator levels (Valeri and Vanderweele, 2013). Based on the mediation test, we used enrichment analysis to explore the potential metabolic pathway through which prenatal pesticide exposure affected birth outcomes. Blood serum metabolites were mapped to biochemical pathways, and metabolomics enrichment analysis was performed using MetaboAnalyst 4.0 (<https://www.metaboanalyst.ca/>). The metabolites map was established using Interactive Pathways Explorer (iPath) platform (<https://pathways.embl.de/>), which showed the pathway feature of intermediated metabolites.

The R software (version 3.6.1 for windows, Vienna, Austria) was used for all statistical analyses. Two-sided $P < 0.05$ was considered to be statistically significant.

3. Results

3.1. Characteristics of study population

The distribution of characteristics of both mothers and infants is shown in Table 1. For mothers, the average maternal age (SD) was 28.05 (2.68) years. There were 70.59% of subjects with normal BMI before pregnancy, and most of mothers (98.04%) were primitive. About one-third of the women's education was undergraduate, and one in five households earned more than 200,000 yuan a year. Very few women reported smoking ($n = 1$, 0.98%) and alcohol consumption ($n = 7$, 6.86%) during pregnancy. One-third of them reported exposure to passive-smoking, and average weight gain during pregnancy was 16.86 kg. For infants, the mean ($\pm$ SD) length of gestation was 39.32 ± 1.23 weeks. The number of preterm birth was relatively low ($n = 4$, 3.92%). The mean ($\pm$ SD) birth weight was 3459.90 ± 480.94 g, and few infants had low birth weight ($n = 3$, 2.94%). For the newborns, 47.08% were male. As the low proportions ($n < 4$, $< 4\%$) of the births were preterm or had low birth weight, we did not conduct the analysis on the outcomes including preterm and low birth weight.

3.2. Distribution of pesticide internal exposure levels

Fig. 1 shows the distributions of internal exposure levels of each pesticide, including the detection rates and pesticide levels. The detection rates of 37 pesticides ranged from 0.98% to 89.22% in the study population. There were 6 pesticides with a detection rate more than 80%, which were simazine, o,p'-DDT, p,p'-DDD, β -HCH, δ -HCH, and α -HCH. Almost all of them were organochlorine pesticides except for simazine. Of them, p,p'-DDD showed the highest exposure levels, up to 7.96 ng/mL (Table S1). There were 19 pesticides with detectable rate $> 20\%$ which were further analyzed for associational study.

Table 1
Characteristics of the study population (n = 102).

Characteristics	n (%) or mean
Maternal characteristics	
Maternal age (year, mean $\pm$ SD)	28.05 $\pm$ 2.68
Pre-pregnancy BMI (kg/m ²)	
< 18.5	24 (23.53)
18.5–23.9	72 (70.59)
≥ 24	6 (5.88)
Parity [No. (%)]	
0	100 (98.04)
≥ 1	2 (1.96)
Education [No. (%)]	
< Bachelor degree	66 (64.71)
$\geq$ Bachelor degree	36 (35.29)
Occupation [No. (%)]	
Manager/Financial staff	23 (22.55)
Education staff	18 (17.65)
Service industry staff	15 (14.71)
Worker	8 (7.84)
Housewife	4 (3.92)
Driver	3 (2.94)
Others	31 (30.39)
Family annual income [No. (%)]	
< 200 thousand yuan	81 (79.41)
≥ 200 thousand yuan	21 (20.59)
Drinking during pregnancy [No. (%)]	
Yes	7 (6.86)
No	95 (93.14)
Smoking during pregnancy [No. (%)]	
Yes	1 (0.98)
No	101 (99.02)
Passive-smoking [No. (%)]	
Yes	34 (33.33)
No	68 (66.67)
Pregnancy weight gain (kg $\pm$ SD)	16.86 $\pm$ 6.18
Newborn characteristics	
Birth weight (g, mean $\pm$ SD)	3459.90 $\pm$ 480.94
< 2500	3 (2.94)
2500–4000	83 (81.37)
≥ 4000	16 (15.69)
Sex [No. (%)]	
Male	47 (47.08)
Female	55 (53.92)
Length of gestation (week, mean $\pm$ SD)	39.32 $\pm$ 1.23
< 37	4 (3.92)
≥ 37	98 (96.08)

Abbreviations: BMI: body mass index.

3.3. Association of pesticide exposure with birth weight and length of gestation

Associations between pesticides with detection rate $> 20\%$ & $< 50\%$ and birth weight and length of gestation using dichotomous (detectable vs. undetectable) exposure values are shown in Fig. S1 and Table S3. Compared to the women with undetectable exposure to mecarbam, the women who exposed to mecarbam tended to have lower birth weight, indicating its possible effect on decreasing birth weight. Fig. S2 and Table S4 showed the associations between pesticides with detection rate $> 50\%$ grouped by tertile and birth outcomes. β -HCH in second and third tertiles showed a significant decline in birth weight compared with the first tertile. And the P value for trend was significant ($P < 0.01$). Similarly, p,p' -DDD and o,p' -DDT in third tertile had significant declines in birth weight compared with the first tertile. And the P value for trend of p,p' -DDD was significant.

Given the fact that our results were significant in the above association studies, we conducted sensitivity analysis by adjusting for different sets of confounding variables, as shown in Fig. 2A. Pesticides with detection rate $> 50\%$ were treated as continuous variables, and those with detection rate $< 50\%$ were treated as dichotomous variables. Detailed results are shown in Table S5. Interestingly, we found a robust association between β -HCH and mecarbam in maternal blood

and birth weight. We found a decrease of 55.14 g in birth weight per ln-unit increase of β -HCH ($\beta_{\text{adjusted}} = -55.14$; $P < 0.01$) (Fig. 2B, Table S5).

The combined effect of mixed pesticides exposure on birth outcomes is shown in Fig. S3, indicating a negative association of overall pesticides on birth weight. The probabilities of inclusion derived from the BKMR models are shown in Table S6, and the PIP of β -HCH was higher than 0.5, which revealed the strongest effect of β -HCH on decreasing birth weight. However, none of the PIPs for length of gestation model was higher than 0.5. Our results were consistent with the results of linear regression models. However, we did not identify any interaction between β -HCH and other pesticide exposure and birth weight (Fig. S4).

3.4. Meta-analysis on the association between level of β -HCH and birth weight

Based on the pesticide exposome wide associational analysis, we found the negative associations of prenatal β -HCH and mecarbam exposure with birth weight. Due to the limited number of published papers on mecarbam, we only conducted a meta-analysis on effect of prenatal β -HCH exposure on birth weight with the inclusion of our current study. Studies matching the eligible criteria were retrieved and the results of literature research and selection progression are shown in Fig. S5. We identified 335 studies including 60 articles from PubMed, 177 from Web of Science, 97 from Embase and 1 from Cochrane Library before December 2019. After the exclusion of duplicated records and studies based on the inclusion criteria, there were 247 records left. Furthermore, after screening the titles and abstracts, 36 records remained for full text reading. Finally, 5 eligible studies with a total of 2497 sample size were included in the meta-analysis. These studies were from USA (Fenster et al., 2006), Australia (Callan et al., 2016), Spain (Lopez-Espinosa et al., 2011), Norway and Sweden (Lauritzen et al., 2017) and China (Fang et al., 2019). General characteristics of the included studies are summarized in Table S7. Quality assessments of included researches are shown in Table S8. The score for each study was greater than seven points, which proved the quality of studies included.

Our data were merged with the above five articles reporting associations between blood β -HCH level and continuous birth weight by means of adjusted β coefficients using meta-analysis. The scale of β -HCH in each study was transformed into ln scale. The results from the Q-test showed a P -value of 0.15 and I^2 -value of 36.12%, indicating that, despite the genetic difference, our current study and the previous studies on the degree of effect on association between the β -HCH level in blood and birth weight showed no heterogeneity. Therefore, fixed-effect model was applied to evaluate such association. As shown in Fig. 3A, the level of β -HCH was significantly associated with birth weight (pooled $\beta = -22.96$; 95%CI, -35.84 to -10.07 ; $I^2 = 36.12\%$). Funnel-plot with regard to Egger's tests ($P = 0.10$) and Begg's Test ($P = 0.56$) showed that there was no evidence of publication bias (Fig. S6). And sensitivity analyses using the "leave-one-out" approach indicated that none of the included studies could affect the robustness of current study (Fig. 3B).

3.5. Mediation analysis on the effect of identified pesticides on birth weight by blood serum metabolomics analysis

A total of 161 metabolites were detected in blood serum samples using metabolomics analysis, including 53 in amino acid metabolism, 47 in lipid metabolism, 18 in carbohydrate metabolism, 14 in nucleotide metabolism and 29 in other metabolisms. After excluding 36 metabolites with the detectable rate less than 40% according to previous report (Xia et al., 2018), 125 urinary metabolites were remained in datasets for further analysis.

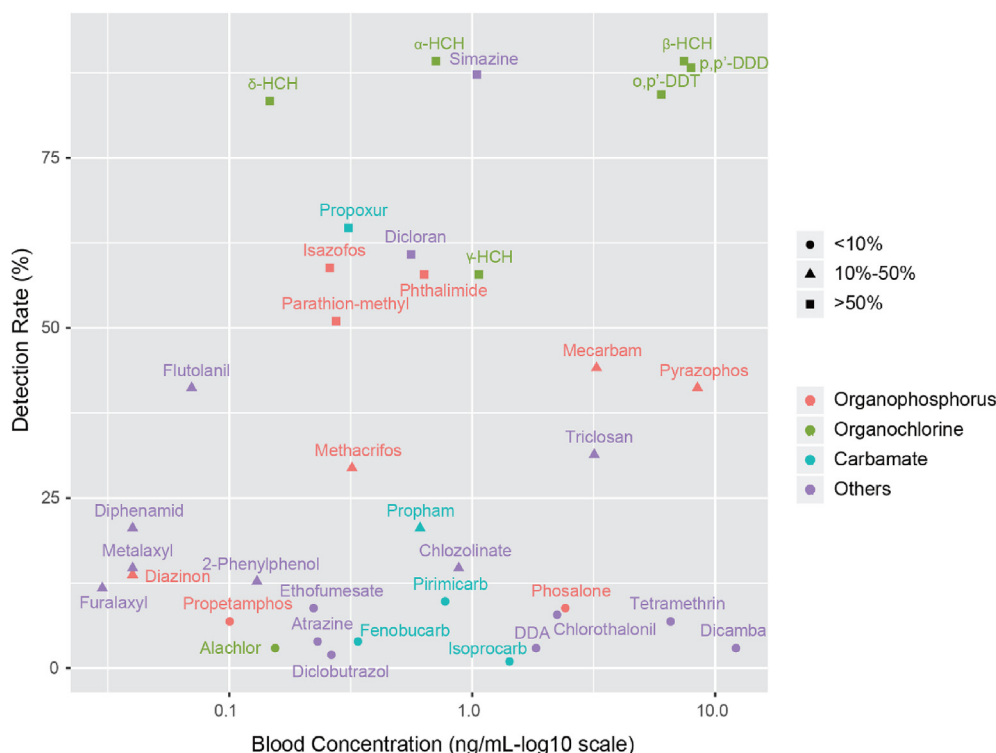


Fig. 1. Concentration map of 37 pesticides in pregnant women's serum. For pesticides with detection rates < 10%, the maximum level is shown. For those with detection rates $\geq 10\%$ & < 50%, the concentration of the 90% level is shown. For those with detection rates $\geq 50\%$, the median level is shown.

3.5.1. The association between blood serum metabolome and β -HCH

Of the 125 metabolites, 21 candidate metabolites showed significant associations with β -HCH (Table S9). Among them, birth weight was positively associated with thyroxine, L-tryptophan and rhamnose (Table S9 & Fig. 4A). Furthermore, we performed mediation analysis to estimate the proportion of the β -HCH exposure effect on birth weight as mediated by the three metabolites (Table 2). The result showed that the

three metabolites were significant as mediators. Adjusted estimates of the proportion of mediation suggested that thyroxine was responsible for 22.88% (indirect effect: -13.99 g; 95%CI: $-32.97, -1.41$) of the association between the β -HCH exposure and birth weight (Table 2). At the same time, L-Tryptophan and rhamnose significantly mediated the association between β -HCH and birth weight responsible for 20.51% and 35.68%, respectively.

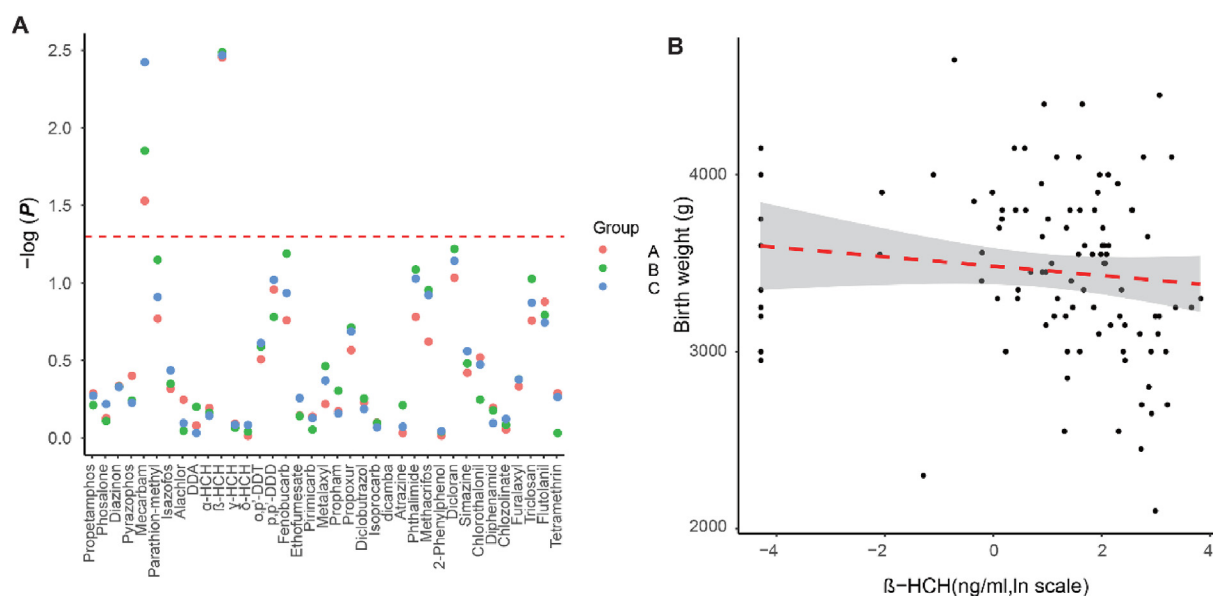


Fig. 2. The associations between pesticides and birth weight. **A:** $\log(P)$ value for multi-linear regression models adjusted for different covariates of relationship between pesticides and birth weight. Model A: adjusted by pre-pregnancy BMI, length of gestation, weight gain, passive smoking and drinking. Model B: adjusted by pre-pregnancy BMI, length of gestation, weight gain, passive smoking, drinking, maternal age, family income and infant sex. Model C: adjusted by pre-pregnancy BMI, length of gestation, weight gain, passive smoking, drinking, maternal age, family income, infant sex and maternal education. The red dotted line indicates the cutoff value ($P < 0.05$, $-\log(P) > 1.3$). **B:** The plot diagram showing the correlation between β -HCH in maternal blood and birth weight. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

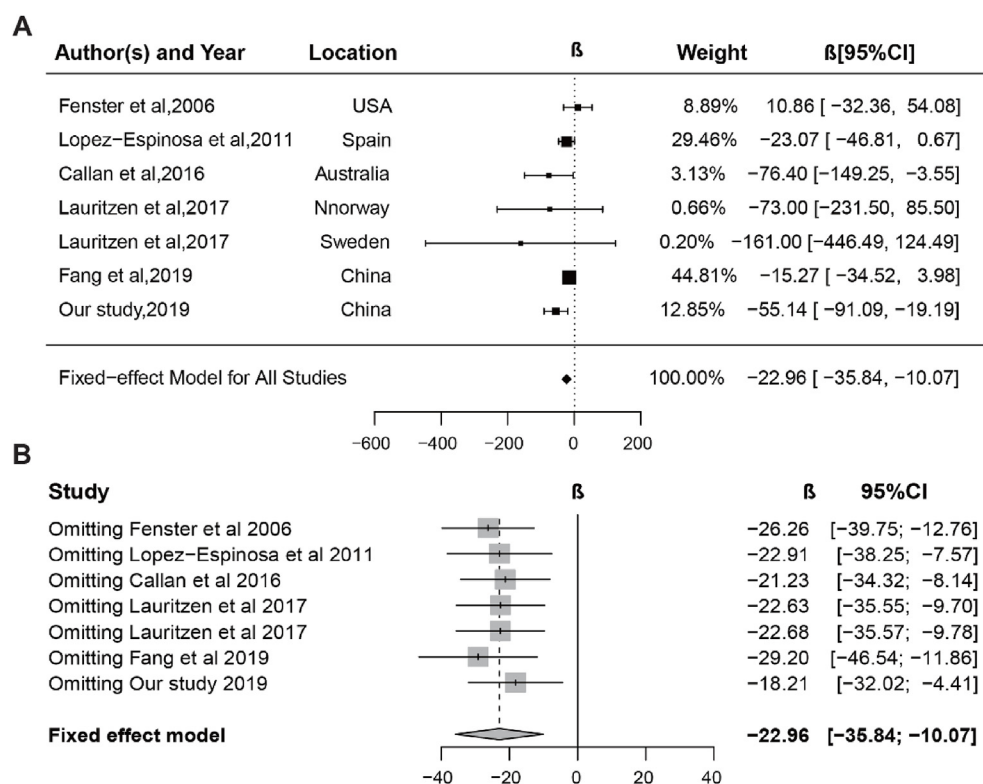


Fig. 3. Meta-analysis of the association between β -HCH and birth weight. **A:** Forest plot of the pooled β with 95% CI for the association of β -HCH and birth weight. The squares and horizontal lines represent the corresponding β s and 95% CIs. The vertical lines show the β of 0. The solid diamond and horizontal line represents the pooled β and 95% CI. **B:** Sensitivity analysis of meta-analysis. The squares and horizontal lines represent the corresponding β s and 95% CIs. The vertical lines show the β of 0. The solid diamond and horizontal line represents the pooled β and 95% CI.

Based on the three metabolites identified, our metabolomics pathway analysis revealed that thyroid hormone synthesis including thyroxine was mostly enriched (Fig. 4B).

3.5.2. The association between blood serum metabolome and mecarbam

There were 19 candidate metabolites found with significant associations with mecarbam (Table S10). Multiple linear analysis showed that birth weight was positively associated with glyceraldehyde, L-aspartyl-L-phenylalanine, L-tryptophan, rhamnose and uridine (Fig. 4C). Similarly, we performed mediation analysis to estimate the proportion of the mecarbam exposure effect on birth weight as mediated by the five metabolites. The result showed that the five metabolites were significant as mediators (Table 2). Adjusted estimates of the proportion of mediation suggested that glyceraldehyde was responsible for 22.88% (indirect effect: -75.80 g; 95%CI: -152.21, -17.09) of the association between the mecarbam exposure and birth weight (Table 2).

Based on these five metabolites identified, we found that glycerolipid metabolism including glyceraldehyde was mostly enriched (Fig. 4C). The 5 metabolites and the enrichment results were indicated in Fig. 4D. The overall process for metabolomics study and related findings were summarized in Fig. S7.

Metabolic network of the mediating metabolites and related metabolic pathways in KEGG general metabolic pathway map are shown in Fig. 4E.

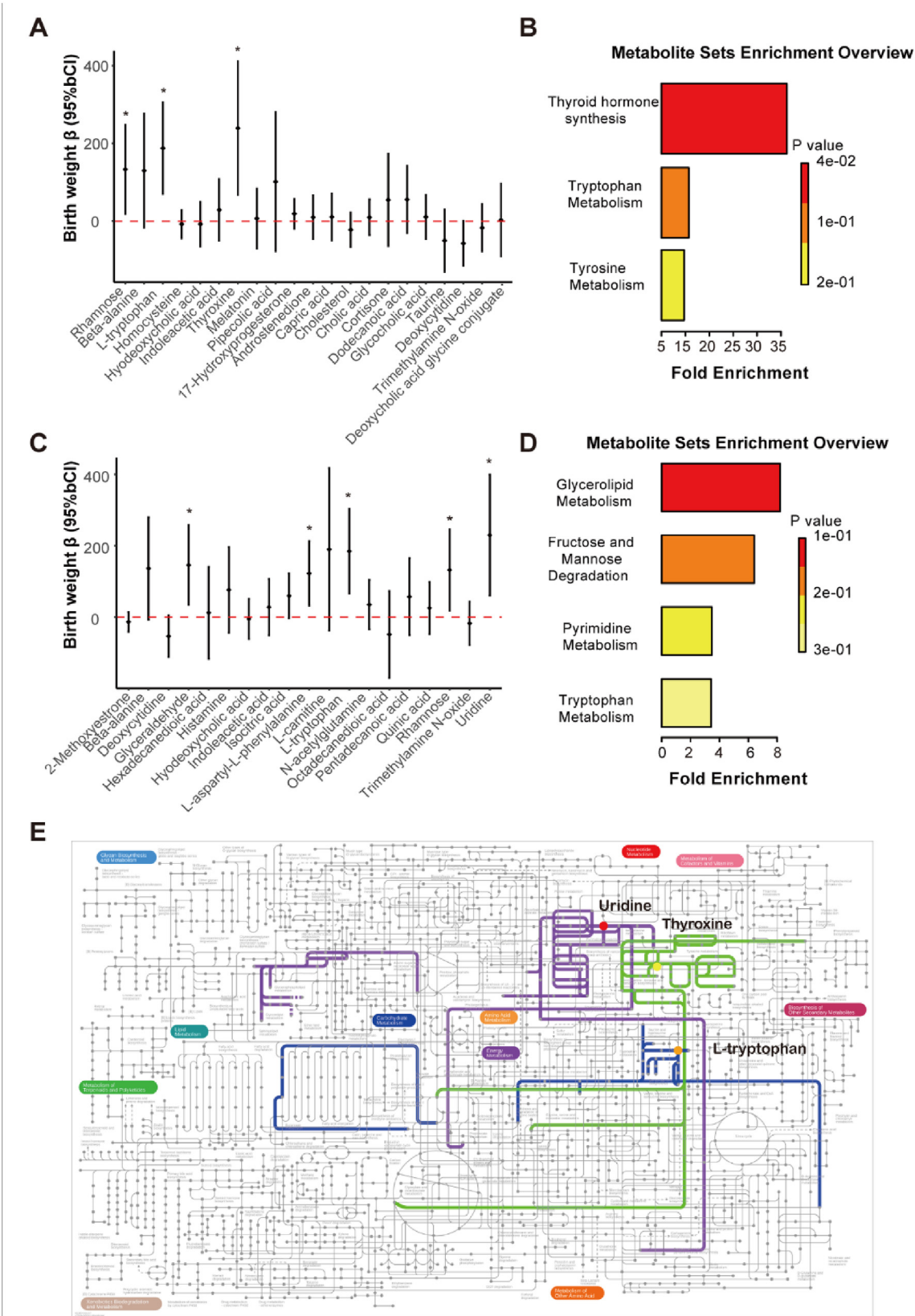
4. Discussion

In this study, based on the exposome analysis covering maternal blood serum levels of 37 pesticides, we found that decreased birth weight was significantly associated with increased levels of mecarbam and β -HCH. Meta-analysis confirmed a significantly negative association between exposure to β -HCH and birth weight. Using metabolomics analysis, the glycerolipid metabolism and thyroid hormone metabolism were found to be the mostly enriched mediating metabolic pathways for the effects of mecarbam and β -HCH, respectively.

Because of the lack of knowledge on pesticide, the overuse of

pesticides is common in China (Zhang et al., 2015). Based on the pesticide exposome analysis in this study, there were 6 pesticides including simazine, o, p'-DDT, and p, p'-DDD, β -HCH, δ -HCH, and α -HCH with a detection rate more than 80% in the serum of participants. Almost all of them were organochlorine pesticides with the property as environmental persistence pollutants, whose toxic effect is a hot topic in the recent environment research (LaKind et al., 2018; Han et al., 2019). HCH is an organochlorine pesticide used as a fungicide. It has several isomers such as α -HCH, β -HCH, γ -HCH, and δ -HCH. β -HCH is the predominant isomer in the environment and human body as a result of its resistance to microbial degradation and strong potential to bioaccumulation (Chen et al., 2009; Feng et al., 2016; Zhang et al., 2017). In China, it is reported that the usage of HCHs and DDTs was 4.9 and 0.4 million tons in 30 years in the last century, respectively, accounting for about 30% and 20% of the global usage respectively (Wei et al., 2007; Qiu et al., 2009; Feng et al., 2016). Today, organochlorine pesticides are still reported to be used in some Asian and African countries (Silvia Quadroni, 2019). The levels of HCHs and DDTs detected in our study are comparable with those reported in other studies in China (Guo et al., 2014), but are significantly higher than those from Mexico, Arabia and India (Waliszewski et al., 2012; Al-Othman et al., 2014; Ruiz-Suarez et al., 2014; Sharma et al., 2015).

As for the significant association between increasing maternal β -HCH serum levels and decreased birth weight, there was evidence that HCHs could have undergone transplacental transfer to produce adverse reproductive outcomes (Zhang et al., 2018; Yin et al., 2019). Conclusions on the relationship between β -HCH and birth weight are still inconsistent (Gladen et al., 2003; Fenster et al., 2006; Pathak et al., 2011; Dewan et al., 2013; Guo et al., 2014; Callan et al., 2016). To address the inconsistency of these studies, results from our meta-analysis showed the significant association between β -HCH and birth weight. The funnel-plot was symmetric, and the association was not modified by single study. Growth retardation related to organochlorine pesticides has been indicated because of genotoxic, immunotoxic, endocrine disruption (anti-progesterone effects), cytotoxic (lipid peroxidation) and fetotoxic effects (placental dysfunction) (Pathak et al., 2010). However,



(caption on next page)

the mechanism underlying the effect of prenatal exposure to HCHs on birth weight has not yet been reported.

Based on our untargeted metabolomics analysis, we discovered that β -HCH was associated with birth weight through disturbing the thyroid hormone metabolism. β -HCH-related metabolite in thyroid hormone metabolism was thyroxine, which was also positively associated with birth weight in our study. Evidence showed that organochlorine pesticides could disrupt the endocrine system, which was involved in fetal

Fig. 4. Metabolomics results on the metabolites associated with β -HCH/mecarbam exposure. Association between log-transformed serum metabolites concentrations (A: significantly associated with β -HCH; C: significantly associated with mecarbam) and birth weight. Error bars indicate 95% confidence intervals. *Indicates P value < 0.05 . Summary of enrichment analysis for the β -HCH-related (B) and mecarbam-related (D) metabolites by using Metaboanalyst. (E) Metabolic network of the mediating metabolites and related metabolic pathways in KEGG general metabolic pathway map. Yellow dots represent the mediating metabolites associated with β -HCH; red dots represent the mediating metabolites associated with mecarbam; the orange dots represent mediating metabolites associated with these two pesticides. Green lines represent pathways associated with mediating metabolites of β -HCH; purple lines represent pathways associated with mediating metabolites of mecarbam; blue line represents tryptophan metabolism, associated with mediating metabolite of these two pesticides. The metabolites and pathways not indicated in the general pathway map are not shown. The original general metabolic pathway map is available at <https://pathways.embl.de/ipath3.cgi>. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

development (Bourguignon and Parent, 2010; Schug et al., 2011). Another study suggested the association between altered thyroid levels during pregnancy and exposure to β -HCH (Alvarez-Pedrerol et al., 2009). Llop et al. found that β -HCH exposure could interact with deiodinase enzymes which are important in the activation and deactivation of thyroid hormones (Llop et al., 2017). In other studies, organochlorines are thought to either to decrease or mimic the thyroid hormone's biological action because of the structural similarity between some organochlorines and thyroid hormone, such as increasing clearance of thyroid hormone through inducing thyroid-metabolizing enzymes or competitive binding to transport proteins (Hood et al., 2003; Purkey et al., 2004). Meanwhile, Medici et al. (Medici et al., 2013) found that maternal FT4 levels in early pregnancy were associated with birth weight. The level of thyroid hormone during pregnancy directly determined the iodine nutrition status (Glinioer, 2004), affecting the fetal growth. Furthermore, it was reported that organochlorine pesticides in maternal blood could transfer through the placenta and affect newborn thyroid hormone levels (Maervoet et al., 2007; Li et al., 2014), which is essential for physical and psycho-neuron-intellectual development in fetuses.

We found that maternal mecarbam exposure exerted a negative effect on birth weight of newborns. So far few studies explored the toxicity on human after chronic exposure to mecarbam. However, in a clinical case study, mecarbam was reported to affect the peripheral system after acute poisoning (Stamboulis E et al., 1991). Another case study found that mecarbam was able to cross the placental barrier and thus posed a threat to fetus (H. TSOUKALI-PAPADOPOULOU, 1987). Based on our metabolomics analysis, we found that mecarbam had a potential decreasing effect on birth weight through disturbing glyceraldehyde, and its related glycerolipid metabolism was mostly enriched. The metabolism of glycerolipid can be perturbed by exposure to the pesticides. It is reported that pesticides affect glycerolipid metabolism through upgrading genes such as HR96 and Dma/PLA2 (Fuertes et al., 2019). Glyceraldehyde, an intermediate of carbohydrate metabolism, was found to be positively associated with birth weight in our study. In addition, glyceraldehyde is a highly reactive compound that can modify and cross-link proteins (Sakasaki-Sakai et al., 2017). Glyceraldehyde-

modified proteins appear to be cytotoxic, depress intracellular glutathione levels, and induce reactive oxygen species (ROS) production (Usui et al., 2004). Indeed, maternal carbohydrate metabolism and ROS are two events related to birth weight (Catalano PM and Amini, 1995; Ishii et al., 2011). To the best of our knowledge, this is the first report about the potential metabolites mediating mecarbam and birth weight.

Another purpose of our study was to explore the association between maternal pesticide exposure and length of gestation. However, no robust significant association was found. Indeed, associations between pesticides and length of gestation differed from studies (Anand et al., 2017). Our study, however, was consistent with most of these research results (Fenster et al., 2006; Pathak et al., 2009).

Our study had the following strengths: first, this study was conducted in two integrated omics perspective, namely pesticide exposome and metabolome analyses, providing the opportunity for data driven and hypothesis free study design and getting a comprehensive understanding of maternal pesticide exposure and birth outcomes. The pesticide exposome analysis was sensitive, stable and accurate. The metabolomics analysis provided novel mechanistic insights into the associations. Second, the samples in our study for pesticide internal exposure measurement and metabolomics analysis were collected before outcome occurred, which provided evidence of causal association. Third, the meta-analysis we conducted covered a total of 2599 pregnant women, which provided human evidence of the largest sample size so far on the topic regarding pesticide exposure and birth weight.

This study also had several limitations. First of all, metabolites of pesticides were not measured in this study, and it is possible that exposure was actually higher for some of the pesticides which are rapidly metabolized. However, some metabolite standards of pesticides are difficult to obtain. Additionally, it is possible that the pesticides share common metabolites in urine with other chemicals, which may compromise the assessment of pesticide exposure (Prapamontol et al., 2014). Thus, blood sample is the optimal option for the high-throughput pesticide detection. More precise pesticide profiling is still required for the pesticides with specific metabolites by including those specific metabolites in the future. Secondly, our study was also limited in spot sample analysis which may introduce exposure

Table 2
Mediation analysis of metabolic molecule.

Metabolic molecule (Ln-scale)	AME		ADE		Proportion (%)
	β (95%CI)	P value	β (95%CI)	P value	
<i>β-HCH</i>					
Thyroxine	-13.99 (-32.97, -1.41)	< 0.01	-40.87 (-72.31, -0.12)	0.06	22.88
L-tryptophan	-11.94 (-23.67, -0.96)	0.02	-46.29 (-73.33, -10.63)	0.02	20.51
Rhamnose	-17.41 (-39.27, -3.94)	0.04	-34.85 (-72.06, 4.64)	0.08	35.68
<i>mecarbam</i>					
Glyceraldehyde	-75.80 (-152.21, -17.09)	< 0.01	-168.26 (-293.88, -15.29)	0.06	30.61
L-aspartyl-L-phenylalanine	-48.14 (-119.5, -7.47)	0.02	-210.84 (-349.67, -82.19)	0.02	17.47
L-tryptophan	-55.50 (-108.43, -11.87)	0.02	-197.05 (-316.89, -56.45)	< 0.01	22.31
Rhamnose	-71.27 (-148.01, -16.88)	0.04	-164.51 (-347.97, 18.93)	0.08	31.02
Uridine	-55.79 (-127.78, -9.45)	0.04	-191.51 (-320.46, -50.82)	0.02	21.76

Abbreviations: AME, average mediation effect; ADE, average direct effect.

The mediation model was adjusted for pre-pregnancy BMI, maternal age, weight gain, drinking, length of gestation, passive-smoking, family income, infant sex and maternal education.

misclassification. However, we included the subjects without living habits changes during pregnancy, and the half-lives of the β -HCH and organochlorine pesticides may be long (Waclawek et al., 2019). In addition, although the risk of misclassification may still exist, since this would be most likely non-differential, one would expect mainly attenuation of observed exposure–effect associations (Mendiola et al., 2010). Therefore, the lack of association of non-persistent pesticides such as organophosphorus pesticides still needs to be further studied.

5. Conclusions

Based on the comprehensive pesticide exposome and metabolome wide associational study combined with meta-analysis, we found that exposure to mecarbam and β -HCH were significantly related to a decrease in birth weight by disturbing glyceraldehyde metabolism and thyroid hormone metabolism, providing new insights into the toxic effects of exposure to pesticides on birth outcomes.

Funding

This work was supported by the National Natural Science Foundation of China (81630085, 81872650); Medical Science and Technology Projects of Jiangsu Province (BE2016633).

Ethical approval

The present study was reviewed and approved by the Institutional Review Board of Nanjing Medical University, China NJMUIRB (2010) 0028.

Declaration of competing interest

None of the authors have a conflict of interest pertaining to this work.

Acknowledgements

None.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.envres.2019.109087>.

References

- Al-Othman, A., Yakout, S., Abd-Alrahman, S.H., Al-Daghri, N.M., 2014. Strong associations between the pesticide hexachlorocyclohexane and type 2 diabetes in Saudi adults. *Int. J. Environ. Res. Public Health* 11 (9), 8984–8995.
- Alvarez-Pedrerol, M.G.M., Ibarluzea, J., Rebagliato, M., Rodriguez, A., Espada, M., Goñi, F., Basterrechea, M., Sunyer, J., 2009. Organochlorine compounds, iodine intake, and thyroid hormone levels during pregnancy. *Environ. Sci. Technol.* 43 (20), 7909–7915.
- Anand, M., Singh, L., Agarwal, P., Saroj, R., Taneja, A., 2017. Pesticides exposure through environment and risk of pre-term birth: a study from Agra city. *Drug Chem. Toxicol.* 1–7.
- Barberini, L., Noto, A., Fattuoni, C., Grapov, D., Casanova, A., Fenu, G., Gaviano, M., Carboni, R., Ottonello, G., Crisafulli, M., Fanos, V., Dessi, A., 2014. Urinary metabolomics (GC-MS) reveals that low and high birth weight infants share elevated in-utero concentrations at birth. *J. Matern. Fetal Neonatal Med.* 27 (Suppl. 2), 20–26.
- Best, K.E., Tennant, P.W.G., Rankin, J., 2017. Survival, by birth weight and gestational age, in individuals with congenital heart disease: a population-based study. *J Am Heart Assoc* 6 (7).
- Bobb, J.F., Valeri, L., Claus Henn, B., Christiani, D.C., Wright, R.O., Mazumdar, M., Godleski, J.J., Coull, B.A., 2015. Bayesian kernel machine regression for estimating the health effects of multi-pollutant mixtures. *Biostatistics* 16 (3), 493–508.
- Bourguignon, J.P., Parent, A.S., 2010. Early homeostatic disturbances of human growth and maturation by endocrine disruptors. *Curr. Opin. Pediatr.* 22 (4), 470–477.
- Bracewell-Milnes, T., Saso, S., Abdalla, H., Nikolau, D., Norman-Taylor, J., Johnson, M., Holmes, E., Thum, M.Y., 2017. Metabolomics as a tool to identify biomarkers to predict and improve outcomes in reproductive medicine: a systematic review. *Hum. Reprod. Update* 23 (6), 723–736.
- Bradman, A., Barr, D.B., Henn, B.G.C., Drumheller, T., Curry, C., Eskenazi, B., 2003. Measurement of pesticides and other toxicants in amniotic fluid as a potential biomarker of prenatal exposure: a validation study. *Environmental Health Perspectives* 111 (14), 1779–1782.
- Callan, A.C., Hinwood, A.L., Heyworth, J., Phi, D.T., Odland, J.O., 2016. Sex specific influence on the relationship between maternal exposures to persistent chemicals and birth outcomes. *Int. J. Hyg Environ. Health* 219 (8), 734–741.
- Catalano PM, D.N., Amini, S.B., 1995. Maternal carbohydrate metabolism and its relationship to fetal growth and body composition. *Am. J. Obstet. Gynecol.* 172 (5).
- Chen, D., Zhang, X., Mai, B., Sun, Q., Song, J., Luo, X., Zeng, E.Y., Hale, R.C., 2009. Polychlorinated biphenyls and organochlorine pesticides in various bird species from northern China. *Environ. Pollut.* 157 (7), 2023–2029.
- Cheng, Q., P.M.D., Lucinda, J., England, Joyce, Martin, A., William, Callaghan, M., 2007. Effects of different data-editing methods on trends in race-specific preterm delivery rates, United States, 1990–2002. *Paediatric and Perinatal Epidemiology*, vol. 21, pp. 41–49.
- Dana, B., Barr, J.R.B., Maggio, Vincent L., Whitehead Jr., Ralph D., Sadowski, Melissa A., Whyatt, Robin M., Needham, Larry L., 2002. A multi-analyte method for the quantification of contemporary. *J. Chromatogr. B* 778, 99–111.
- David Moher, L.S., Clarke, Mike, Ghersi, Davina, Liberati, Alessandro, Petticrew, Mark, Paul, Shekelle, Stewart, Lesley A., PRISMA-P Group, 2015. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Syst. Rev.* 349 g7647-7625.
- Dewan, P., Jain, V., Gupta, P., Banerjee, B.D., 2013. Organochlorine pesticide residues in maternal blood, cord blood, placenta, and breastmilk and their relation to birth size. *Chemosphere* 90 (5), 1704–1710.
- EPA, 2017. from. <https://www.epa.gov/pesticides/pesticides-industry-sales-and-usage-2008-2012-market-estimates>.
- Fang, J., Liu, H., Zhao, H., Wong, M., Xu, S., Cai, Z., 2019. Association of prenatal exposure to organochlorine pesticides and birth size. *Sci. Total Environ.* 654, 678–683.
- Faul, F., Erdfelder, E., Lang, A.G., Buchner, A., 2007. G*Power 3: a flexible statistical power analysis program for the social, behavioral, and biomedical sciences. *Behav. Res. Methods* 39 (2), 175–191.
- Favretto, D., Cosmi, E., Ragazzi, E., Visentin, S., Tucci, M., Fais, P., Cecchetto, G., Zanardo, V., Viel, G., Ferrara, S.D., 2012. Cord blood metabolomic profiling in intrauterine growth restriction. *Anal. Bioanal. Chem.* 402 (3), 1109–1121.
- Feng, J., Hu, P., Zhang, F., Sun, J., 2016. HCHs and DDTs in Yellow River of Henan section—a typical agricultural area in China: levels, distributions and risks. *Environ. Geochem. Health* 38 (6), 1241–1253.
- Fenster, L., Eskenazi, B., Anderson, M., Bradman, A., Harley, K., Hernandez, H., Hubbard, A., Barr, D.B., 2006. Association of in utero organochlorine pesticide exposure and fetal growth and length of gestation in an agricultural population. *Environmental Health Perspectives* 114 (4), 597–602.
- Fuertes, I., Jordao, R., Pina, B., Barata, C., 2019. Time-dependent transcriptomic responses of *Daphnia magna* exposed to metabolic disruptors that enhanced storage lipid accumulation. *Environ. Pollut.* 249, 99–108.
- Gladden, B.C., Shkiryak-Nyzhnyk, Z.A., Chyslovska, N., Zadorozhnaja, T.D., Little, R.E., 2003. Persistent organochlorine compounds and birth weight. *Ann. Epidemiol.* 13 (3), 151–157.
- Glinier, D., 2004. The regulation of thyroid function during normal pregnancy: importance of the iodine nutrition status. *Best Pract. Res. Clin. Endocrinol. Metabol.* 18 (2), 133–152.
- Guo, H., Jin, Y., Cheng, Y., Leaderer, B., Lin, S., Holford, T.R., Qiu, J., Zhang, Y., Shi, K., Zhu, Y., Niu, J., Bassig, B.A., Xu, S., Zhang, B., Li, Y., Hu, X., Chen, Q., Zheng, T., 2014. Prenatal exposure to organochlorine pesticides and infant birth weight in China. *Chemosphere* 110, 1–7.
- TSOUKALI-PAPADOPOULOU, S.N.H., 1987. Mother-fetus postmortem toxicologic analysis in a fatal overdose with mecarbam. *Forensic Sci. Int.* 35, 249–252.
- Han, X., Meng, L., Li, Y., Li, A., Turyk, M.E., Yang, R., Wang, P., Xiao, K., Li, W., Zhao, J., Zhang, Q., Jiang, G., 2019. Associations between exposure to persistent organic pollutants and thyroid function in a case-control study of east China. *Environ. Sci. Technol.* 53 (16), 9866–9875.
- Hood, A., Allen, M.L., Liu, Y., Liu, J., Klaassen, C.D., 2003. Induction of T4 UDP-GT activity, serum thyroid stimulating hormone, and thyroid follicular cell proliferation in mice treated with microsomal enzyme inducers. *Toxicol. Appl. Pharmacol.* 188 (1), 6–13.
- Hornung, R.W., Reed, L.D., 1990. Estimation of average concentration in the presence of nondetectable values. *Appl. Occup. Environ. Hyg* 5 (1), 46–51.
- Ishii, T., Miyazawa, M., Onodera, A., Yasuda, K., Kawabe, N., Kirinashizawa, M., Yoshimura, S., Maruyama, N., Hartman, P.S., Ishii, N., 2011. Mitochondrial reactive oxygen species generation by the SDHC V69E mutation causes low birth weight and neonatal growth retardation. *Mitochondrion* 11 (1), 155–165.
- Julian PT Higgins, S.G., 2011. *Cochrane Handbook for Systematic Reviews of Interventions*. from. <http://handbook-5-1.cochrane.org/>.
- Jusko, T.A., van den Dries, M.A., Pronk, A., Shaw, P.A., Guxens, M., Spaan, S., Jaddoe, V.W., Tiemeier, H., Longnecker, M.P., 2019. Organophosphate pesticide metabolite concentrations in urine during pregnancy and offspring nonverbal IQ at age 6 years. *Environ. Health Perspect.* 127 (1), 17007.
- LaKind, J.S., Lehmann, G.M., Davis, M.H., Hines, E.P., Marchitti, S.A., Alcalá, C., Lorber, M., 2018. Infant dietary exposures to environmental chemicals and infant/child health: a critical assessment of the literature. *Environ. Health Perspect.* 126 (9), 96002.
- Lauritzen, H.B., Larose, T.L., Oien, T., Sandanger, T.M., Odland, J.O., van de Bor, M., Jacobsen, G.W., 2017. Maternal serum levels of perfluoroalkyl substances and organochlorines and indices of fetal growth: a Scandinavian case-cohort study. *Pediatr.*

- Res. 81 (1–1), 33–42.
- Lawlor, D.A., Ronalds, G., Clark, H., Smith, G.D., Leon, D.A., 2005. Birth weight is inversely associated with incident coronary heart disease and stroke among individuals born in the 1950s: findings from the Aberdeen Children of the 1950s prospective cohort study. *Circulation* 112 (10), 1414–1418.
- Li, C., Cheng, Y., Tang, Q., Lin, S., Li, Y., Hu, X., Nian, J., Gu, H., Lu, Y., Tang, H., Dai, S., Zhang, H., Jin, C., Zhang, H., Jin, Y., Jin, Y., 2014. The association between prenatal exposure to organochlorine pesticides and thyroid hormone levels in newborns in Yancheng, China. *Environ. Res.* 129, 47–51.
- Llop, S., Murcia, M., Alvarez-Pedrerol, M., Grimalt, J.O., Santa-Marina, L., Julvez, J., Goni-Irigoyen, F., Espada, M., Ballester, F., Rebagliato, M., Lopez-Espinosa, M.J., 2017. Association between exposure to organochlorine compounds and maternal thyroid status: role of the iodothyronine deiodinase 1 gene. *Environ. Int.* 104, 83–90.
- Lopez-Espinosa, M.J., Murcia, M., Iniguez, C., Vizcaino, E., Llop, S., Vioque, J., Grimalt, J.O., Rebagliato, M., Ballester, F., 2011. Prenatal exposure to organochlorine compounds and birth size. *Pediatrics* 128 (1), e127–134.
- Luzardo, O.P., Mahtani, V., Troyano, J.M., Alvarez de la Rosa, M., Padilla-Perez, A.I., Zumbado, M., Almeida, M., Burillo-Putze, G., Boada, C., Boada, L.D., 2009. Determinants of organochlorine levels detectable in the amniotic fluid of women from Tenerife Island (Canary Islands, Spain). *Environ. Res.* 109 (5), 607–613.
- Maervoet, J., Vermeir, G., Covaci, A., Van Larebeke, N., Koppen, G., Schoeters, G., Nelen, V., Baeyens, W., Schepens, P., Viaene, M.K., 2007. Association of thyroid hormone concentrations with levels of organochlorine compounds in cord blood of neonates. *Environ. Health Perspect.* 115 (12), 1780–1786.
- Medici, M., Timmermans, S., Visser, W., de Muinck Keizer-Schrama, S.M., Jaddoe, V.W., Hofman, A., Hooijkaas, H., de Rijke, Y.B., Tiemeier, H., Bongers-Schokking, J.J., Visser, T.J., Peeters, R.P., Steegers, E.A., 2013. Maternal thyroid hormone parameters during early pregnancy and birth weight: the Generation R Study. *J. Clin. Endocrinol. Metab.* 98 (1), 59–66.
- Melsen, W.G., Bootsma, M.C., Rovers, M.M., Bonten, M.J., 2014. The effects of clinical and statistical heterogeneity on the predictive values of results from meta-analyses. *Clin. Microbiol. Infect.* 20 (2), 123–129.
- Mendiola, J., Jorgensen, N., Andersson, A.M., Calafat, A.M., Ye, X., Redmon, J.B., Drobnis, E.Z., Wang, C., Sparks, A., Thurston, S.W., Liu, F., Swan, S.H., 2010. Are environmental levels of bisphenol A associated with reproductive function in fertile men? *Environ. Health Perspect.* 118 (9), 1286–1291.
- Pathak, R., Ahmed, R.S., Tripathi, A.K., Guleria, K., Sharma, C.S., Makhijani, S.D., Banerjee, B.D., 2009. Maternal and cord blood levels of organochlorine pesticides: association with preterm labor. *Clin. Biochem.* 42 (7–8), 746–749.
- Pathak, R., Mustafa, M., Ahmed, R.S., Tripathi, A.K., Guleria, K., Banerjee, B.D., 2010. Association between recurrent miscarriages and organochlorine pesticide levels. *Clin. Biochem.* 43 (1–2), 131–135.
- Pathak, R., Mustafa, M.D., Ahmed, T., Ahmed, R.S., Tripathi, A.K., Guleria, K., Banerjee, B.D., 2011. Intra uterine growth retardation: association with organochlorine pesticide residue levels and oxidative stress markers. *Reprod. Toxicol.* 31 (4), 534–539.
- Perera, F.P., Rauh, V., Tsai, W.Y., Kinney, P., Camann, D., Barr, D., Bernert, T., Garfinkel, R., Tu, Y.H., Diaz, D., Dietrich, J., Whyatt, R.M., 2003. Effects of transplacental exposure to environmental pollutants on birth outcomes in a multiethnic population. *Environ. Health Perspect.* 111 (2), 201–205.
- Prapamontol, T., Sutan, K., Laoyang, S., Hongsiabong, S., Lee, G., Yano, Y., Hunter, R.E., Ryan, P.B., Barr, D.B., Panuwet, P., 2014. Cross validation of gas chromatography-flame photometric detection and gas chromatography-mass spectrometry methods for measuring dialkylphosphate metabolites of organophosphate pesticides in human urine. *Int. J. Hyg Environ. Health* 217 (4–5), 554–566.
- Purkey, H.E., Palaninathan, S.K., Kent, K.C., Smith, C., Safe, S.H., Sacchettini, J.C., Kelly, J.W., 2004. Hydroxylated polychlorinated biphenyls selectively bind transthyretin in blood and inhibit amyloidogenesis: rationalizing rodent PCB toxicity. *Chem Biol* 11 (12), 1719–1728.
- Qiu, Y.-W., Zhang, G., Guo, L.-L., Cheng, H.-R., Wang, W.-X., Li, X.-D., Wai, O.W.H., 2009. Current status and historical trends of organochlorine pesticides in the ecosystem of Deep Bay, South China. *Estuar. Coast Shelf Sci.* 85 (2), 265–272.
- Rauch, S.A., Braun, J.M., Barr, D.B., Calafat, A.M., Khoury, J., Montesano, A.M., Yoltou, K., Lanphear, B.P., 2012. Associations of prenatal exposure to organophosphate pesticide metabolites with gestational age and birth weight. *Environ. Health Perspect.* 120 (7), 1055–1060.
- Robinson, O., Keski-Rahkonen, P., Chatzi, L., Kogevinas, M., Nawrot, T., Pizzi, C., Plusquin, M., Richiardi, L., Robinot, N., Sunyer, J., Vermeulen, R., Vrijheid, M., Vineis, P., Scalbert, A., Chadeau-Hyam, M., 2018. Cord blood metabolic signatures of birth weight: a population-based study. *J. Proteome Res.* 17 (3), 1235–1247.
- Ruiz-Suarez, L.E., Castro-Chan, R.A., Rivero-Perez, N.E., Trejo-Acevedo, A., Guillen-Navarro, G.K., Geissen, V., Bello-Mendoza, R., 2014. Levels of organochlorine pesticides in blood plasma from residents of malaria-endemic communities in Chiapas, Mexico. *Int. J. Environ. Res. Public Health* 11 (10), 10444–10460.
- Sakasai-Sakai, A., Takata, T., Takino, J.I., Takeuchi, M., 2017. Impact of intracellular glyceraldehyde-derived advanced glycation end-products on human hepatocyte cell death. *Sci. Rep.* 7 (1), 14282.
- Schug, T.T., Janesick, A., Blumberg, B., Heindel, J.J., 2011. Endocrine disrupting chemicals and disease susceptibility. *J. Steroid Biochem. Mol. Biol.* 127 (3–5), 204–215.
- Sharma, A., Gill, J.P., Bedi, J.S., 2015. Monitoring of pesticide residues in human blood from Punjab, India. *Bull. Environ. Contam. Toxicol.* 94 (5), 640–646.
- Shuqin, J., Fang, Z., 2018. Zero growth of chemical fertilizer and pesticide use: China's objectives, progress and challenges. *Journal of Resources and Ecology* 9 (sp1), 50–58.
- Silvia Quadroni, R.B., 2019. An unnoticed issue: organochlorine pesticides in tobacco products around the world. *An Unnoticed Issue: Organochlorine Pesticides in Tobacco Products Around the World*, vol. 219. pp. 54–57.
- Stamboulis E, P.A., Vassilopoulos, D., Davaki, P., Manta, P., KapakiE, 1991. Neuropathy following acute intoxication with Mecarbam (OP ester). *Acta Neurol. Scand.* 83, 198–200.
- Stang, A., 2010. Critical evaluation of the Newcastle-Ottawa scale for the assessment of the quality of nonrandomized studies in meta-analyses. *Eur. J. Epidemiol.* 25 (9), 603–605.
- Tang, R., Chen, M., Zhou, K., Chen, D., Yu, J., Hu, W., Song, L., Hang, B., Wang, X., Xia, Y., 2015. Prenatal lignan exposures, pregnancy urine estrogen profiles and birth outcomes. *Environ. Pollut.* 205, 261–268.
- Toichuev, R.M., Zhilova, L.V., Paizildaev, T.R., Khametova, M.S., Rakhmatillaev, A., Sakibaev, K.S., Madykova, Z.A., Toichueva, A.U., Schlumpf, M., Weber, R., Lichtensteiger, W., 2018 Nov. Organochlorine pesticides in placenta in Kyrgyzstan and the effect on pregnancy, childbirth, and newborn health. *Environ. Sci. Pollut. Res. Int.* 25 (32), 31885–31894.
- Usui, T., Shizuuchi, S., Watanabe, H., Hayase, F., 2004. Cytotoxicity and oxidative stress induced by the glyceraldehyde-related Maillard reaction products for HL-60 cells. *Biosci. Biotechnol. Biochem.* 68 (2), 333–340.
- Valeri, L., Vanderweele, T.J., 2013. Mediation analysis allowing for exposure-mediator interactions and causal interpretation: theoretical assumptions and implementation with SAS and SPSS macros. *Psychol. Methods* 18 (2), 137–150.
- Waclawek, S., Silvestri, D., Hrabak, P., Padil, V.V.T., Torres-Mendieta, R., Waclawek, M., Cernik, M., Dionysiou, D.D., 2019. Chemical oxidation and reduction of hexachlorocyclohexanes: a review. *Water Res.* 162, 302–319.
- Waliszewski, S.M., Caba, M., Herrero-Mercado, M., Saldariaga-Norena, H., Meza, E., Zepeda, R., Martinez-Valenzuela, C., Gomez Arroyo, S., Villalobos Pietrini, R., 2012. Organochlorine pesticide residue levels in blood serum of inhabitants from Veracruz, Mexico. *Environ. Monit. Assess.* 184 (9), 5613–5621.
- Wang, P., Tian, Y., Wang, X.J., Gao, Y., Shi, R., Wang, G.Q., Hu, G.H., Shen, X.M., 2012. Organophosphate pesticide exposure and perinatal outcomes in Shanghai, China. *Environ. Int.* 42, 100–104.
- Wei, D., Kameya, T., Urano, K., 2007. Environmental management of pesticidal POPs in China: past, present and future. *Environ. Int.* 33 (7), 894–902.
- Whyatt, R.M., Rauh, V., Barr, D.B., Camann, D.E., Andrews, H.F., Garfinkel, R., Hoepner, L.A., Diaz, D., Dietrich, J., Reyes, A., Tang, D., Kinney, P.L., Perera, F.P., 2004. Prenatal insecticide exposures and birth weight and length among an urban minority cohort. *Environmental Health Perspectives* 112 (10), 1125–1132.
- Xia, B., Zhu, Q., Zhao, Y., Ge, W., Zhao, Y., Song, Q., Zhou, Y., Shi, H., Zhang, Y., 2018. Phthalate exposure and childhood overweight and obesity: urinary metabolomic evidence. *Environ. Int.* 121 (Pt 1), 159–168.
- Yin, S., Zhang, J., Guo, F., Zhao, L., Poma, G., Covaci, A., Liu, W., 2019. Transplacental transfer of organochlorine pesticides: concentration ratio and chiral properties. *Environ. Int.* 130, 104939.
- Zhang, C., Hu, R., Shi, G., Jin, Y., Robson, M.G., Huang, X., 2015. Overuse or underuse? An observation of pesticide use in China. *Sci. Total Environ.* 538, 1–6.
- Zhang, Q., Xia, Z., Wu, M., Wang, L., Yang, H., 2017. Human health risk assessment of DDTs and HCHs through dietary exposure in Nanjing, China. *Chemosphere* 177, 211–216.
- Zhang, X., Wu, X., Lei, B., Jing, Y., Jiang, Z., Zhang, X., Fang, X., Yu, Y., 2018. Transplacental transfer characteristics of organochlorine pesticides in paired maternal and cord sera, and placentas and possible influencing factors. *Environ. Pollut.* 233, 446–454.
- Zhou, K., Ding, X., Yang, J., Hu, Y., Song, Y., Chen, M., Sun, R., Dong, T., Xu, B., Han, X., Wu, K., Zhang, X., Wang, X., Xia, Y., 2018. Metabolomics reveals metabolic changes caused by low-dose 4-tert-octylphenol in mice liver. *Int. J. Environ. Res. Public Health* 15 (12), 2686.